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CLASS : P5
ROLL : 17

C – PROGRAMMING WEEK - 11

CODE

P1.c > main()

```c
#include <stdio.h>
#include <string.h>

struct Student {
    int id;
    char name[50];
    int m1, m2, m3;
};

void analyzeStudent(struct Student s);

int main() {
    struct Student s1;

    printf("Enter Student ID: ");
    scanf("%d", &s1.id);

    printf("Enter Student Name: ");
    getchar();
    scanf("%[^\n]", s1.name);

    printf("Enter marks in 3 subjects: ");
    scanf("%d %d %d", &s1.m1, &s1.m2, &s1.m3);

    analyzeStudent(s1);

    return 0;
}

void analyzeStudent(struct Student s) {
    float total = s.m1 + s.m2 + s.m3;
    float percentage = total / 3.0;
    char result[10];
    char grade;

    if (s.m1 < 35 || s.m2 < 35 || s.m3 < 35) {
        strcpy(result, "Fail");
        grade = 'F';
    } else {
        strcpy(result, "Pass");

        if (percentage >= 90) {
            grade = 'A';
        } else if (percentage >= 75 && percentage < 90) {
            grade = 'B';
        } else if (percentage >= 60 && percentage < 75) {
            grade = 'C';
        } else if (percentage >= 50 && percentage < 60) {
            grade = 'D';
        } else {
            grade = 'E';
        }
    }

    printf("Student Details\n");
    printf("ID : %d\n", s.id);
    printf("Name : %s\n", s.name);
    printf("Marks : %.2f %.2f %.2f\n", (float)s.m1, (float)s.m2, (float)s.m3);
    printf("Total : %.2f\n", total);
    printf("Percentage : %.2f\n", percentage);
    printf("Result : %s\n", result);

    if(strcmp(result, "Pass") == 0) {
        printf("Grade : %c\n", grade);
    }
}
```

PROGRAM - 1

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-11> cd "c:\User
 ($?) { gcc P1.c -o P1 } ; if ($?) { .\P1 }
Enter Student ID: 101
Enter Student Name: Rahul
Enter marks in 3 subjects: 85 78 92
Student Details
ID : 101
Name : Rahul
Marks : 85.00 78.00 92.00
Total : 255.00
Percentage : 85.00
Result : Pass
Grade : B
PS C:\Users\Chethan S\Documents\WEEK-11>
```

CODE

```c
//PES1UG25AM095
#include <stdio.h>
#include <string.h>

struct Consumer {
    int cno;
    char name[50];
    float units;
};

void calculateBill(int cno, char name[], float units);

int main() {
    struct Consumer c1;

    printf("Enter Consumer Number: ");
    scanf("%d", &c1.cno);

    printf("Enter Consumer Name: ");
    getchar();
    scanf("%[^\n]", c1.name);

    printf("Enter Units Consumed: ");
    scanf("%f", &c1.units);

    calculateBill(c1.cno, c1.name, c1.units);

    return 0;
}

void calculateBill(int cno, char name[], float units) {
    float bill_amt = 0;
    float surcharge = 0;
    float final_bill = 0;

    if (units <= 100) {
        bill_amt = units * 2.00;
    } else if (units <= 200) {
        bill_amt = (100 * 2.00) + ((units - 100) * 3.50);
    } else {
        bill_amt= (100 * 2.00) + (100 * 3.50) + ((units - 200) * 5.00);
    }

    if (bill_amt > 1000) {
        surcharge = bill_amt * 0.10;
    } else {
        surcharge = 0;
    }

    final_bill = bill_amt + surcharge;

    printf("\nElectricity Bill Details\n");
    printf("Consumer Number : %d\n", cno);
    printf("Consumer Name   : %s\n", name);
    printf("Units Consumed  : %.2f\n", units);
    printf("Bill Amount     : %.2f\n", bill_amt);
    printf("Surcharge       : %.2f\n", surcharge);
    printf("Final Bill      : %.2f\n", final_bill);
}
```

PROGRAM - 2

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-11> cd "c:\
 ($?) { gcc P2.c -o P2 } ; if ($?) { .\P2 }
Enter Consumer Number: 207
Enter Consumer Name: CHETHAN
Enter Units Consumed: 250

Electricity Bill Details
Consumer Number : 207
Consumer Name   : CHETHAN
Units Consumed  : 250.00
Bill Amount     : 800.00
Surcharge       : 0.00
Final Bill      : 800.00
PS C:\Users\Chethan S\Documents\WEEK-11>
```

PROGRAM - 3

CODE

```c
//PES1UG25AM095
#include<stdio.h>

struct Expense
{
float tpp;
int nop;
float hotel_charge;
float discount;
};

struct Booking
{
int booking_id;
char name[30];
char destination[30];
struct Expense exp;
};


int main()
{
int n;

printf("Enter number of bookings: ");
scanf("%d",&n);

struct Booking b[n];

for(int i=0;i<n;i++)
{
printf("\n\nEnter details of booking %d\n",i+1);

printf("Enter Booking ID: ");
scanf("%d",&b[i].booking_id);

printf("Enter Customer name: ");
scanf("%s",&b[i].name);

printf("Enter Destination: ");
scanf("%s",&b[i].destination);

printf("Enter Travel Fare per person: ");
scanf("%f",&b[i].exp.tpp);

printf("Enter Number of person: ");
scanf("%d",&b[i].exp.nop);

printf("Enter Hotel Charges: ");
scanf("%f",&b[i].exp.hotel_charge);

printf("Enter Discount percentage: ");
scanf("%f",&b[i].exp.discount);

}

float discount_amt[n];

printf("\n\n============Travel Booking Details==============\n\n");

for(int i=0;i<n;i++)
{
float total_fare = b[i].exp.tpp  * b[i].exp.nop;

float total_before_disount = total_fare + b[i].exp.hotel_charge;

float discounted_amt = (total_before_disount * (b[i].exp.discount/100) );
discount_amt[i] = discounted_amt;

float final_cost = total_before_disount - discounted_amt;

printf("\n\nBooking %d\n",i+1);

printf("Booking ID  : %d\n",b[i].booking_id);

printf("Customer name   : %s\n",b[i].name);

printf("Destination : %s\n",b[i].destination);

printf("Fare per person : %.2f\n",b[i].exp.tpp);

printf("Number of person    : %d\n",b[i].exp.nop);

printf("Hotel Charges   : %.2f\n",b[i].exp.hotel_charge);

printf("Discount percentage : %.2f\n",b[i].exp.discount);

printf("Total Travel Fare   : %.2f\n",total_fare);

printf("Total cost Before Discount  : %.2f\n",total_before_disount);

printf("Discount Amount     : %.2f\n",discount_amt[i]);

printf("Final Tour Cost     : %.2f\n",final_cost);

if(final_cost > 50000)
    printf("Booking Category    : Premium Booking");
else
    printf("Booking Category    : Regular Booking");

}

printf("\n\n=========Customers with Discount Amount Greater Than Rs.5000=========\n");
for(int i=0;i<n;i++)
{
if(discount_amt[i] > 5000)
    {printf("Customer name  : %s\n",b[i].name);
    printf("Discount Amount     : %.2f\n",discount_amt[i]);}
}

return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-11> cd "c:\Users\Chethan S\Documents\
 ($?) { gcc P3.c -o P3 } ; if ($?) { .\P3 }
Enter number of bookings: 3


Enter details of booking 1
Enter Booking ID: 1
Enter Customer name: Chethan
Enter Destination: Goa
Enter Travel Fare per person: 8000
Enter Number of person: 4
Enter Hotel Charges: 15000
Enter Discount percentage: 10


Enter details of booking 2
Enter Booking ID: 102
Enter Customer name: Chandru
Enter Destination: Kashmir
Enter Travel Fare per person: 12000
Enter Number of person: 5
Enter Hotel Charges: 18000
Enter Discount percentage: 12


Enter details of booking 3
Enter Booking ID: 103
Enter Customer name: Bhavith
Enter Destination: Mangluru
Enter Travel Fare per person: 5000
Enter Number of person: 3
Enter Hotel Charges: 10000
Enter Discount percentage: 5


============Travel Booking Details=====


Booking 1
Booking ID      : 1
Customer name   : Chethan
Destination     : Goa
Fare per person : 8000.00
Number of person        : 4
Hotel Charges   : 15000.00
Discount percentage     : 10.00
Total Travel Fare       : 32000.00
Total cost Before Discount      : 47000.00
Discount Amount         : 4700.00
Final Tour Cost         : 42300.00
Booking Category        : Regular Booking

Booking 2
Booking ID      : 102
Customer name   : Chandru
Destination     : Kashmir
Fare per person : 12000.00
Number of person        : 5
Hotel Charges   : 18000.00
Discount percentage     : 12.00
Total Travel Fare       : 60000.00
Total cost Before Discount      : 78000.00
Discount Amount         : 9360.00
Final Tour Cost         : 68640.00
Booking Category        : Premium Booking

Booking 3
Booking ID      : 103
Customer name   : Bhavith
Destination     : Mangluru
Fare per person : 5000.00
Number of person        : 3
Hotel Charges   : 10000.00
Discount percentage     : 5.00
Total Travel Fare       : 15000.00
Total cost Before Discount      : 25000.00
Discount Amount         : 1250.00
Final Tour Cost         : 23750.00
Booking Category        : Regular Booking

=========Customers with Discount Amount Greater Than Rs.5000=========
Customer name   : Chandru
Discount Amount         : 9360.00
PS C:\Users\Chethan S\Documents\WEEK-11>
```

PROGRAM - 4

CODE

```c
//PES1UG25AM095
#include<stdio.h>

struct Employee
{
int emp_id;
char name[20];
int projects;
float performance_score;
};

int main()
{
int n;

printf("Enter number of Employees: ");
scanf("%d",&n);

struct Employee emp[n];

struct Employee *ptr = emp;

for(int i=0;i<n;i++)
{
printf("\n\nEnter details of Employee %d\n",i+1);

printf("Enter Employee ID: ");
scanf("%d",&(ptr + i)->emp_id);

printf("Enter Employee name: ");
scanf("%s",(ptr + i)->name);

printf("Enter Number of projects completed: ");
scanf("%d",&(ptr + i)->projects);

printf("Enter Performance Score: ");
scanf("%f",&(ptr + i)->performance_score);

}

printf("\n\n============Employee Details==============\n\n");

for(int i=0;i<n;i++)
{

printf("Employee %d\n",i+1);

printf("Employee ID : %d\n",(ptr + i)->emp_id);

printf("Employee name    : %s\n",(ptr + i)->name);

printf("Projects Completed  : %d\n",(ptr + i)->projects);

printf("Performance Score   : %.2f\n",(ptr + i)->performance_score);

}

printf("\n\n=================Employees with More than 5 projects================\n");
for(int i=0;i<n;i++)
{
if((ptr + i)->projects > 5)
    {
    printf("Employee ID: %d , ",(ptr + i)->emp_id);
    printf("Employee name: %s , ",(ptr + i)->name);
    printf("Projects Completed: %d\n",(ptr + i)->projects);
    }

}
printf("\n\n=================Employees with Performance Score >=8.0================\n");
for(int i=0;i<n;i++)
{
if((ptr + i)->performance_score > 8)
    {
    printf("Employee ID: %d , ",(ptr + i)->emp_id);
    printf("Employee name: %s , ",(ptr + i)->name);
    printf("Projects Completed: %.2f\n",(ptr + i)->performance_score);
    }

}

}
```

OUTPUT

```
Enter number of Employees: 3

Enter details of Employee 1
Enter Employee ID: 101
Enter Employee name: Rahul
Enter Number of projects completed: 6
Enter Performance Score: 8.5

Enter details of Employee 2
Enter Employee ID: 102
Enter Employee name: Sneha
Enter Number of projects completed: 4
Enter Performance Score: 7.8

Enter details of Employee 3
Enter Employee ID: 103
Enter Employee name: Arjun
Enter Number of projects completed: 8
Enter Performance Score: 9.1


============Employee Details==============

Employee 1
Employee ID            : 101
Employee name          : Rahul
Projects Completed     : 6
Performance Score      : 8.50

Employee 2
Employee ID            : 102
Employee name          : Sneha
Projects Completed     : 4
Performance Score      : 7.80

Employee 3
Employee ID            : 103
Employee name          : Arjun
Projects Completed     : 8
Performance Score      : 9.10


=================Employees with More than 5 projects================

Employee ID: 101, Employee name: Rahul, Projects Completed: 6
Employee ID: 103, Employee name: Arjun, Projects Completed: 8


=================Employees with Performance Score >= 8.0================

Employee ID: 101, Employee name: Rahul, Performance Score: 8.50
Employee ID: 103, Employee name: Arjun, Performance Score: 9.10
```